The header search bar on HP's sites utilizes a **federated search architecture**. Interacting with the header search bar generates up to **5 distinct XHR/Fetch network requests** across three distinct backend services (HawkSearch, HP CSS Knowledge API, and Analytics engines).

1. Complete Inventory of Header Search XHR Requests

Request 1: HawkSearch Autocomplete API (Suggestions & Editorial Content)

- **Trigger:** User types ≥2 characters in the search input (debounced by ~250ms).
- **Method:** `POST`
- **URL:** `[https://hp.searchapi-na.hawksearch.com/api/v2/autocomplete](https://hp.searchapi-na.hawksearch.com/api/v2/autocomplete)`
- **Purpose:** Retrieves query completions, popular terms, category facets, and editorial newsroom articles matching the typed string.
- **Request Headers:**
  - `Content-Type: application/json`
  - `X-HawkSearch-IgnoreTracking: false`
- **Request Payload:**
- JSON

```
None
{
  "ClientGuid": "00000000000000000000000000000000",
  "Keyword": "develop",
  "FacetFilters": {},
  "MaxResults": 5
}
```

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- **Response Payload Structure:**
- JSON

```
None
{
  "Count": 42,
  "Categories": [{"Value": "Laptops", "Url":
"/us-en/shop/laptops"}],
```

```
  "Popular": [{"Value": "developer workstation", "Url":
"/search?q=developer+workstation"}],
  "Content": [
    {
      "Value": "HP Expands Local AI Era for Developers",
      "Url": "/us-en/newsroom/press-releases/local-ai.html",
      "Thumb": { "Url": "/images/thumb.jpg" }
    }
  ]
}
```

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Request 2: HP CSS Support Knowledge Base API (Hardware Models & SKUs)

- **Trigger:** Executed in parallel with Request 1 when the user types in the input bar.
- **Method:** `GET`
- **URL:** `[https://css.api.hp.com/knowledge/v2/typeahead/search](https://css.api.hp.com/knowledge/v2/typeahead/search)`
- **Purpose:** Fetches exact product series, model numbers, and part identifiers from HP's core hardware catalog.
- **Query Parameters Breakdown:**
    - `q=develop`: Active search query string.
    - `resultLimit=3`: Restricts hardware result matches to 3 to optimize drop-down height.
    - `languageCode=en`: User region language code.
    - `store=tmsstore`: Specific storefront instance ID.
    - `printFields=productid,productname,seofriendlyname`: Payload field trimming (returns only minimal required attributes).
    - `filters=class:(pm_series_value OR pm_name_value OR pm_number_value)`: Restricts scope to Product Series, Product Name, and Model Part Numbers.
    - `excludeDescendants=true`: Excludes child part variants, returning only top parent product lines.
- **Response Payload Structure:**
- JSON

None

```
{
  "status": "success",
  "results": [
    {
      "productid": "21008543",
      "productname": "HP ZBook Studio G8 Developer Edition",
      "seofriendlyname": "hp-zbook-studio-g8-developer-edition"
    }
  ]
}
```

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Request 3: HawkSearch Telemetry & Event Tracking API

- **Trigger:** User clicks an auto-suggest item, selects a popular keyword, or clicks "View All Results".
- **Method:** `POST`
- **URL:** `[https://hp.tracking-na.hawksearch.com/api/trackevent](https://hp.tracking-na.hawksearch.com/api/trackevent)`
- **Purpose:** Captures click-through rates, term popularity, and autocomplete selection analytics for backend ranking optimization.
- **Request Payload:**
- JSON

None

```
{
  "EventType": "autocomplete_click",
  "EventData": {
    "Keyword": "develop",
    "SelectedTerm": "HP ZBook Studio G8",
    "SelectedUrl": "/us-en/shop/pdp/zbook-g8",
    "SearchGuid": "a1b2c3d4-e5f6-7890-abcd-1234567890ab"
  }
}
```

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Request 4: Adobe Analytics / Omniture Telemetry Beacon

- **Trigger:** Fired on search bar open, search form submission, or zero-results impression.
- **Method:** `GET` or `POST`
- **URL:** `[https://hp.sc.omtrdc.net/b/ss/hpcomglobal/1/JS-n/](https://hp.sc.omtrdc.net/b/ss/hpcomglobal/1/JS-n/)` (or `[https://metrics.hp.com/](https://metrics.hp.com/)`)
- **Purpose:** Tracks site-wide user analytics, conversion funnels, and search usage patterns in Adobe Analytics.
- **Key Parameters:**
  - `v12`: Search Term (`eVar12=develop`)
  - `pe`: `lnk_o` (Custom Link Track)
  - `pev2`: `Header_Search_Submit`
  - `events`: `event15` (Search Initiated)

Request 5: HawkSearch Full Search Results API (For In-Page Search Refinement)

- **Trigger:** Submitted when full search execution occurs without triggering a hard browser redirect (e.g., executing a dynamic search on a search page or overlay).
- **Method:** `POST`
- **URL:** `[https://hp.searchapi-na.hawksearch.com/api/v2/search](https://hp.searchapi-na.hawksearch.com/api/v2/search)`
- **Purpose:** Pulls full, paginated search results along with full facet/filter sets.
- **Request Payload:**
- JSON

```
None
{
  "Keyword": "develop",
  "PageNo": 1,
  "MaxPerPage": 12,
  "SortBy": "score",
  "FacetFilters": {}
}
```

-

2. Request Lifecycle Matrix across User Scenarios

Scenario	Request 1 (HawkSearch Autocomplete)	Request 2 (HP CSS KB Search)	Request 3 (HawkSearch Tracking)	Request 4 (Adobe Analytics)	Request 5 (HawkSearch Full Search)
1. Focus Input Field	❌	❌	❌	❌	❌
2. Typing Query ("develop")	✅ POST	✅ GET	❌	❌	❌
3. Click Suggestion Link	❌	❌	✅ POST	✅ POST	❌ (Redirects Page)
4. Press "Enter" / Full Submit	❌	❌	✅ POST	✅ POST	✅ POST (If AJAX SRP)
5. Clear Input Field	❌ (Aborts active)	❌ (Aborts active)	❌	❌	❌